

Assignment 3

Due: April 1, 2026 at 11:59 pm (Central Time)

Q1. Consider the training set shown in Table 1:

A	B	C	Label
0	0	0	+
0	0	1	-
0	1	0	-
0	1	1	-
1	0	0	+
1	0	1	-
1	1	0	-
1	1	1	+

Table 1. Training set (A, B, and C are input features)

- (a) (30%) Estimate the following probabilities: $P(A=1|+)$, $P(A=0|+)$, $P(B=1|+)$, $P(B=0|+)$, $P(C=1|+)$, $P(C=0|+)$, $P(A=1|-)$, $P(A=0|-)$, $P(B=1|-)$, $P(B=0|-)$, $P(C=1|-)$, $P(C=0|-)$, $P(-)$, $P(+)$.
- (b) (20%) Use the estimates of probabilities in part (a) to predict the class label for a test sample ($A=1$, $B=0$, $C=0$) using the naïve Bayes approach. Please provide the details of computations.

Q2. (30%) Write a Python code to train a Support-Vector Machine model (with arbitrary hyperparameters) on the training set shown in Table 1 and return the model's prediction for a test sample ($A=0$, $B=1$, $C=0$). Both the Python code and result need to be included in the submission.

Q3. (20%) What weights (i.e., W_1 and W_2) and bias (i.e., b) values can enable a Perceptron model to work the same as the NOR logical operation (shown in Table 2)?

X_1	X_2	NOR (X_1, X_2)
0	0	1
0	1	0
1	0	0
1	1	0

Table 2. The NOR logical operation